

SYSTEM FLOW & SCOPE — FOR THE RECORD

LoveTap.Me

Complete approved flow, module breakdown, scope classification, acceptance criteria, and launch checklist.
Updated 30 July 2026.

1. System Overview

LoveTap turns a physical NFC pendant into a one-tap tipping instrument. Each pendant carries a permanent web address. When tapped, the phone opens it, the backend reads the product code, and shows an activation screen if unclaimed, or the owner's tip page if claimed. No NFC 'sync', no app required for tippers, and card data never touches LoveTap servers.

Surfaces

- **Web Portal** — `app.lovetap.me` — workers activate & manage; customers tip. Launch-critical and self-sufficient.
- **Admin Panel** — `adminpanel.lovetap.me` — operator manages products, workers, reseller verification, fees, transactions, reconciliation.
- **Mobile App** — optional companion; not a launch dependency.
- **Backend + Square** — Square holds funds, processes cards, and pays workers directly.

2. Complete Flow

2.1 Product identity (permanent, immutable)

- One permanent URL + code per pendant: `app.lovetap.me/t/LT-XXXXXX`.
- Six random characters, case-insensitive, excluding confusables (0/O/1/I); unique in DB; a separate internal UUID is the real key.
- Route and DB accept variable-length codes (6–12) so future batches can be longer without breaking existing pendants.
- Code and URL never change once programmed. Code lookups are rate-limited.

2.2 Product lifecycle

Generated → Programmed/Inventory → Claimable → Assigned → Active

- Only **Claimable** products can be activated — so a pendant on a retail display can't be claimed before purchase.
- Online orders become Claimable automatically on fulfillment/shipping; admin can mark Claimable for in-person sales.
- **Active** = assigned AND the worker has a valid Square connection AND neither product nor worker is disabled/suspended.
- If assigned but not Active, tapping shows “This LoveTap is not currently accepting tips” — not the activation page.

2.3 Worker self-enrollment & claim (store-sellable model)

- 1 Pre-loaded batches of unclaimed codes ship inside the product packaging.
- 2 Buyer taps her new pendant → opens `app.lovetap.me/t/LT-XXXXXX`.
- 3 Unassigned & Claimable → “Activate this LoveTap — LT-XXXXXX”.
- 4 She signs up or logs in on the web; the code is carried through automatically (manual entry is only a fallback).
- 5 Signup collects: first & last name, public display name, mobile, email, TikTok, Instagram, required Terms + Privacy acceptance, optional SMS consent (off by default), User Guide link, consent version + timestamp.
- 6 “Add this LoveTap to your account?” → Confirm → assigned.

Atomic claiming: assignment is atomic — if two people attempt to claim the same code at the same time, only one succeeds, and a product may have only one current ownership assignment at any time.

2.4 One account, multiple LoveTaps

- A returning worker taps a new pendant → signs in → confirms → it appears under “My LoveTaps”.
- Additional products inherit the existing profile and Square connection — no new account, no second OAuth.

2.5 Square connection & reseller verification (separate)

- Worker connects her own Square via OAuth. Permissions: PAYMENTS_WRITE, PAYMENTS_READ, PAYMENTS_WRITE_ADDITIONAL_RECIPIENTS, PAYOUTS_READ (plus profile/location read).
- Intake asks: no account / created via LoveTap link / existing elsewhere → new users routed through the LoveTap referral link; outside accounts flagged (intake info, not proof).
- Two statuses shown: Square: Connected · Reseller: Pending.
- Reseller verification is anchored on the Square merchant ID matched against the SIP roster — not self-report. Stored: merchant ID, location ID, claimed path, status, source, verified-by, date, notes.
- Admin sets: Verified / Not enrolled / Exception approved / Suspended. Payments allowed while pending; no auto-suspend.
- Tokens: encrypted, server-side only, auto-refreshed every 7 days or less, alerts on failure. Worker has Reconnect and Disconnect (disconnect revokes access and stops tips immediately).
- Two revenue streams tracked separately: LoveTap application fee vs reseller residual.

2.6 Customer tipping

- 1 Customer taps an Active pendant → tip page shows the worker's display name.
- 2 Picks \$5 / \$10 / \$20 / \$50 or a custom amount.
- 3 Sees the breakdown first: Tip + LoveTap service fee = Total.
- 4 Enters card on Square's secure form (LoveTap never sees card numbers; HTTPS + required security headers).
- 5 Confirmation + receipt option (receipt available for over \$15) + refund-policy disclosure.

2.7 The money (worker nets the exact tip)

Total charged = (Tip + Square fixed fee + LoveTap fee) ÷ (1 – Square %). Then: Total – Square fee – LoveTap fee = the displayed tip.

- “LoveTap fee” = the actual app_fee_money paid to LoveTap.
- All amounts computed in integer cents with documented rounding; Square's percentage and fixed fee are configurable, not hardcoded.
- Exact-net holds under configured Square pricing; custom merchant pricing and international cards (Square may add ~1.5%) are documented and handled.
- Proven with production reconciliation for \$5, \$10, \$20, \$50, a custom amount and a refund — showing actual Square fee, LoveTap fee, and exact worker net.
- Money settles into the worker's own Square account.

2.8 Refunds, disputes & integrity

- Unique idempotency key per payment attempt; verified Square webhook signatures; duplicate-event protection.
- Refund and dispute statuses tracked; on refunds Square keeps its processing fee and reverses the app fee proportionally.
- A written policy defines who absorbs retained/chargeback costs (also reflected in worker Terms). Permanent transaction and audit history.

2.9 Payouts (Square-controlled)

- LoveTap does not initiate or process payouts. Square pays the worker from her own balance on her Square schedule.
- Worker sees: Square payouts — sent / paid / failed + estimated arrival.
- Admin: view & reconcile Square payout status; flag failed/delayed and direct the worker to Square.

2.10 Reassignment (history preserved)

- Reassigning closes the prior assignment and creates a new one without overwriting history.
- Previous transactions stay with the original worker; only future tips route to the new worker.
- Records administrator, reason, date and time. Code never changes; nothing is deleted. Admin can also disable, transfer, or retire a product.

2.11 What each surface shows

- **Worker (web):** My LoveTaps · Dashboard · Transactions · Square status (Reconnect/Disconnect) · Profile.
- **Admin:** Customers · Products & reseller verification · NFC chips (batches, tap URLs, reassign/disable/retire) · Transactions & audit · Fees (service fee + platform cut, configurable) · Square payout reconciliation.

2.12 Edge cases handled

- Not-yet-Claimable → can't activate. Assigned but not Active → "not currently accepting tips".
- Square disconnected → tips stop immediately. Disabled/lost → inactive without rewriting the chip.
- No NFC phone → type the code printed in the packaging (fallback). Older URLs → still resolve.

3. Modules by Surface

Every module across the four surfaces and the shared backend.

3.1 Web App — Worker Portal (app.lovetap.me, logged in)

- **Authentication** — sign up, log in, forgot/reset password, session handling.
- **Activation / Claim** — tap-to-activate an unclaimed chip; code carried through automatically; “Add this LoveTap to your account” confirmation; manual code entry fallback.
- **My LoveTaps** — list of owned chips with each chip's status (Assigned / Active / Disabled), its tap URL, and an “add another” action.
- **Square connection** — connect, live status (Connected + Reseller Pending/Verified), Reconnect, and Disconnect/Revoke.
- **Dashboard** — total tips, key stats, tips-over-time chart, recent activity.
- **Transactions** — full history with statuses, filters, per-transaction detail, receipt reference.
- **Payouts (Square)** — sent / paid / failed transfers and estimated arrival, read from Square.
- **Profile** — name, public display name, phone, email, TikTok, Instagram, consent preferences, change password, log out.

3.2 Web App — Public Tip Page (app.lovetap.me/t/CODE, no login)

- **Code lookup & routing** — reads the code from the URL and routes: unclaimed → activation, assigned+active → tip page, inactive/disabled → clear message.
- **Recipient display** — shows the worker's public display name.
- **Amount selection** — presets (\$5/\$10/\$20/\$50) + custom amount.
- **Fee breakdown** — Tip + LoveTap service fee = Total, shown before payment.
- **Secure card checkout** — Square Web Payments SDK; card tokenized in-browser; server never sees card data.
- **Result & receipt** — “Thank you” confirmation, receipt option (over \$15), refund-policy disclosure.

3.3 Admin Panel (adminpanel.lovetap.me)

- **Authentication** — admin login (separate from workers).
- **Dashboard** — totals: tips, active workers, fees collected, platform revenue; recent activity.
- **Workers / Customers** — list, search, detail (tips, payouts, chips, Square status, reseller status), activate / deactivate / suspend.
- **Products / NFC Chips** — batch generate, CSV export (codes + tap URLs), lifecycle management (Inventory / Claimable / Assigned / Active), mark Claimable, link/assign, reassign (with history), disable / transfer / retire, filters, and stats.
- **Reseller Verification** — merchant-ID matching against the SIP roster, status workflow (Verified / Not enrolled / Exception approved / Suspended), stored fields (merchant ID, location ID, claimed path, source, verified-by, date, notes).
- **Transactions** — all tips across workers; search; detail; refund/dispute status; reference IDs.
- **Payout Reconciliation** — view Square payout status; flag failed/delayed; direct workers to Square.
- **Fees** — customer service fee (% , flat), platform (application) fee, configurable Square rates, live preview.
- **Settings** — admin profile, email-notification preferences, referral-link configuration.
- **Audit Log** — every admin action (reassign, disable, verification change) with who / when / why.

3.4 Mobile App (optional companion)

- Mirrors the worker portal: dashboard, transactions, My LoveTaps, Square status, profile.
- In-app tipping fallback (tap the chip or enter a code, then pay).
- Not a launch dependency; the web portal performs every essential function.

3.5 Backend / API & Integrations

- **Auth service** — worker and admin authentication, tokens/sessions.

- **Chip / code service** — batch generation, unique codes, lookup, lifecycle transitions, atomic claiming, rate-limiting.
- **Payments service** — quote/create-intent, charge, application fee, gross-up math, idempotency.
- **Square OAuth service** — connect, callback, encrypted token storage, auto-refresh (≤ 7 days), revoke/disconnect.
- **Webhooks** — signature verification, duplicate-event protection, refund/dispute events.
- **Payout reconciliation** — reads Square Payouts API for status.
- **Reseller verification service** — merchant-ID vs SIP roster.
- **Notifications/alerts** — OAuth-refresh failures and operational alerts.
- **Data models** — Worker, Product/Chip, Assignment history, Transaction, Payout, FeeConfig, AdminUser, Consent records; permanent audit trail.

4. Scope Classification

The system architecture and acceptance flow are approved as the intended LoveTap build specification. Approval of the architecture is not, by itself, acceptance of any commercial classification or authorization of additional charges. The work below is grouped into three categories.

4.1 Category 1 — Original scope, still to be completed

Agreed original deliverables that remain unbuilt or partially built. These are unfinished original work.

- Web worker self-enrollment: sign up / log in on the web.
- Tap-to-claim activation during sign-up; “Add to my account”; unclaimed → activation and assigned → tip-page routing.
- One account owning multiple claimed codes (“My LoveTaps” basics).
- The core gross-up calculation so the worker receives the full displayed tip.
- Immutable chip URLs and the basic assignment lifecycle (unclaimed / assigned / active; programmed tags locked read-only).
- Atomic claiming and the single-current-owner constraint.

4.2 Category 2 — Corrections & baseline security (to make original features work)

Fixes and baseline requirements needed for the original features to function correctly and safely.

- Removing LoveTap “process / retry payout” functionality — a correction of behaviour that contradicts “Square pays workers directly.”
- Square webhook-signature verification and duplicate-event protection — baseline payment security.
- Secure encrypted token storage and continued authorization (token refresh) — inherent to a functioning Square OAuth implementation.
- Correcting the existing User and Admin guides so they accurately describe the agreed system.

4.3 Category 3 — Genuinely new enhancements

Work beyond the original specification. Mixed items are shown split into their original/baseline part (Category 1 or 2) and their incremental part (Category 3).

- **Reseller verification system** — Square merchant-ID matching to the SIP roster, status workflow, and admin tooling. (Fully new.)
- **Fee configuration extras** — configurable custom Square pricing, international-card treatment, and the expanded production-test matrix. (Core net-tip calculation is Category 1; these are incremental.)
- **OAuth tooling extras** — refresh-failure alerts, expanded Reconnect/Disconnect interfaces, and additional admin tooling. (Secure storage + refresh are Category 2; these are incremental.)
- **Dispute management & audit reporting** — expanded dispute handling and reporting. (Signature verification + duplicate protection are Category 2; these are incremental.)
- **Claimable inventory gating, fulfillment automation, and expanded reassignment history.** (The basic lifecycle is Category 1; these are incremental.)
- **Square payout-reconciliation dashboard.** (Removing process/retry is Category 2; the dashboard is incremental.)
- **Expanded signup fields & consent versioning** (TikTok, Instagram, display name, consent version/timestamps) and **new documentation** beyond correcting the existing guides.

4.4 Outside the codebase (business)

- Reseller residual program (managed under the reseller agreement, separate from the platform fee).
- Written category-level approval from SIP/Square before enrolling adult-entertainment-oriented workers.
- Production Square credentials to move off test mode.

5. Current Build Status

Live now

- Permanent /t/CODE URL and code lookup; tip page for an assigned worker.
- Square OAuth (with additional-recipients permission); direct-to-seller charge with app_fee_money, customer_initiated, idempotency key.
- Card tokenized by Square (server never sees card numbers); configurable platform fee in admin.
- Batch generation + CSV tap URLs; admin panel; app./adminpanel. subdomains with hostname routing; branding.

Still to be completed / built

- Web worker sign-up/login screen + tap-to-activate claim flow + “My LoveTaps” (Category 1).
- Core gross-up net-tip calculation; atomic claiming (Category 1).
- Payout correction; webhook verification + dedupe; token refresh; guide corrections (Category 2).
- Reseller verification; fee/OAuth/dispute extras; inventory gating + fulfillment automation; payout dashboard; expanded signup/consent + new docs (Category 3).

6. Acceptance Criteria (approved)

- 1 Active status:** Active only when assigned + valid Square + not disabled/suspended; assigned-but-inactive shows “not currently accepting tips”.
- 2 Reassignment:** closes prior assignment, keeps prior transactions, routes future tips to the new worker, logs admin/reason/date/time.
- 3 Fee calculation:** LoveTap fee = actual app_fee_money; integer cents with documented rounding; Square % and fixed fee configurable.
- 4 Fee exceptions:** exact-net under configured pricing; custom merchant pricing and international cards documented; reconciliation shows actual Square fee, LoveTap fee, worker proceeds.
- 5 OAuth permissions:** PAYMENTS_WRITE, PAYMENTS_READ, PAYMENTS_WRITE_ADDITIONAL_RECIPIENTS, PAYOUTS_READ.
- 6 Disconnect Square:** workers can disconnect/revoke; a disconnected account immediately stops accepting tips.
- 7 Inventory activation:** online orders auto-Claimable on fulfillment/shipping; admin can mark Claimable for in-person sales.
- 8 Documentation:** replace the existing User and Admin guides before launch.
- 9 Atomic claiming:** assignment is atomic; simultaneous claim attempts result in exactly one successful claimant; only one current ownership assignment may exist per product.

7. Staging Demonstration — Six Journeys

- 1** New worker activates her first pendant.
- 2** Returning worker adds a second product.
- 3** Customer completes a \$5 tip.
- 4** Square is disconnected and tipping becomes unavailable.
- 5** A product is reassigned without losing history.
- 6** A payment is refunded and all three financial pieces reconcile.